

Pathology & Microbiology

(Group A Pathology & Group B Microbiology)

Departmental Objectives

At the end of the course, the students should be able to:

- Describe causes & pathogenesis of diseases
- Demonstrate knowledge and understanding of micro-organisms & their role in disease
- Demonstrate knowledge and understanding of immunology for prevention of diseases
- Describe Haematological disease and Haematological disorders
- Interpret clinical pathological specimen to evaluate the diagnosis of disease

List of Competencies to acquire:

- Demonstrate knowledge and understanding of general pathological, microbiological & preventive knowledge about diseases
- Interpret the pathological, microbiological & serological laboratory test for diagnosis of diseases
- Apply pathological & microbiological knowledge in practicing of dental surgery to improve the oral & dental health in the country & abroad

Distribution of teaching - learning hours

Lecture	Tutorial	Practical	Total Teaching hours	Integrated teaching (Common)	Formative Exam		Summative exam	
					Preparatory leave	Exam time	Preparatory leave	Exam time
116	102	22	240	10hrs	10 days	20days	10 days	25days

Teaching - learning methods, teaching aids and evaluation

Teaching Methods			Teaching aids	In course evaluation
Large group	Small group	Self learning		
Lecture Integrated teaching	Tutorial Practical Demonstration	Self-study & self-assessment	Computer/laptop & Multimedia OHP, Transparency & Transparency marker White board & different colour white board markers Black board & white and coloured chalks Cardavers, prosected parts, bones, viscera Slide and slide projector Microscope	Item Examination Card final Examination written/oral+ practical) Term final (written, oral+ practical)

Related Equipments: Flip Chart, Photograph, Model, Diagram, Preserved specimens

Professional Examination:

Marks distribution of Pathology (G-A) & Microbiology (G-B)

Total marks – 300

- **Written** = 100 (SAQ=70 (G-A 35+ G-B 35) + MCQ =20 (G-A 10+ G-B 10) + Formative Assessment =10 (G-A 05+G-B 05)
- **SOE** = 100 (G-A 50+G-B 50)
- **Practical** = 100 (G-A 50+G-B 50)

110

Learning Objectives	Contents	Teaching Hours
Student will be able to : - State the causes of diseases & Pathology - State the causes of cell injury, types with examples. - State reversible and irreversible cell injury, Necrosis – types and morphology. Apoptosis, Fate, Pathogenic calcification - Identify gross specimen & microscopic examples of necrosis.	1. Cell injury: - Causes and types of cell injury - Reversible and irreversible cell injury - Necrosis 2. Inflammation : - Acute and chronic inflammation - Causative agents, cardinal sign of inflammation - Classification and structure of granuloma - Basic tissue changes in inflammation - Chemical mediators, compliment system. - Fate of different types of inflammation. - Common inflammatory lesions of oral cavity.	Lecture-4 hrs
Student will be able to : - Define acute and chronic inflammation-granulomatous inflammations. - Mention causative agents, cardinal sign of inflammation, - Classify and mention typical structure of granuloma. - State basic tissue changes in inflammation, acute and chronic, Utilities & drawback of inflammatory responses. - Mention morphological types of inflammation, chemical mediators complement system. - State fate of different types of inflammation. - State common inflammatory lesions of oral cavity. - Identify acute, chronic and granulomatous inflammation.		L-6 hrs

Learning Objectives	Contents	Teaching Hours
Student will be able to : ● Define healing, repair and regeneration, angiogenesis ● Outline of Primary and secondary cutaneous wound healing ● State stages of wound healing, Factors influencing wound healing, ● Mention complication and demonstrate understanding of healing after dental surgery (tooth extraction) ● Identity granulation tissue under microscope.	3. Wound healing : ● Definition of healing, repair and regeneration, angiogenesis ● Primary and secondary cutaneous wound healing ● Stages of wound healing, ● Factors influencing wound healing, ● Complication of dental surgery (tooth extraction) ● Healing after dental surgery (tooth extraction)	L-4 hrs
Student will be able to : ● Define oedema and types, aetiopathology of oedema(localized & ● generalized), mechanism of generalized oedema. ● Define thrombosis, embolism and infarction. Mention their formations & fate. ● Define and mention types of shock, mechanism of septic shock, stages of shock. ● Mention body's response to acute hemorrhage. Define Hyperemia and congestion with example.	4. Hemodynamic disorders ● Oedema and types, aetiopathology of oedema(localized & generalized), ● Mechanism of generalized oedema. ● Thrombosis, embolism and infarction, their formations & fate. ● Types of shock, mechanism of septic shock, stages of shock. ● Body's response to acute hemorrhage. Hyperemia and congestion.	L-7 hrs

Learning Objectives	Contents	Teaching Hours
Student will be able to : ● Define atrophy, hyperplasia, hypertrophy, metaplasia with examples. ● Identify gross and microscopic example of adaptive changes. Student will be able to : ● Define Neoplasia, mention nomenclature, features of benign & malignant neoplasia with difference. ● Mention Biological behavior of Neoplasia, anaplasia, invasion, metastasis, carcinogen, para-neoplastic syndrome, precancerous conditions. ● Mention common tumours & tumour like lesions of the oral cavity. ● State different diagnostic procedure of tumour. ● State principles of tissue preservation and procedure of sending a biopsy specimen to the laboratory and cytology. Student will be able to : ● Define common terms used in medical genetics – examples, Cytogenetic disorders, single gene disorders, molecular basis disorders, Diagnosis of genetic disease (Barr body). ● State Pathogenesis of commonly encountered genetic diseases of oral cavity.	5. Adaptive changes: Atrophy, hyperplasia, hypertrophy, metaplasia with examples. 6. Neoplasia (tumour) : ● Neoplasia, Nomenclature, features of benign & malignant neoplasia with difference, ● Biological behavior of Neoplasia, anaplasia, invasion, metastasis, carcinogen, para-neoplastic syndrome, precancerous conditions. ● Common tumours & tumour like lesions of the oral cavity. ● Different diagnostic procedure of tumour. ● Principles of tissue preservation and procedure of sending a biopsy specimen to the laboratory and cytology 7. Genetics: ● Cytogenetic disorders, single gene disorders, molecular basis disorders, Diagnosis of genetic disease (Barr body). ● Pathogenesis of commonly encountered genetic diseases of oral cavity.	L-1 hr L-8 hrs L-1 hr Total tutorial-20hrs Total practical--3 hrs

B- Hematology

Learning Objectives	Contents	Teaching Hours
Student will be able to : ● State introduction of haematology: Haemopoiesis, ● Describe collection of blood for haematological investigation, anticoagulant and blood film staining ● Define Anaemia, and state classification & aetiology of anaemia, Common anaemias: Iron deficiency anaemia, megaloblastic, pancytopenia & aplastic anemia, Haemolytic anaemia: Thalassaemia & Haemoglobinopathies. ● Mention Laboratory investigations of anaemia including haemolytic anaemia ● Describe white blood cell disorders: Leucocytosis, Leucopenia, lymphocytosis and leucomoid blood picture. ● State -classification, aetiology, clinical features and laboratory diagnosis of leukaemia (Acute and Chronic). ● State Multiple myeloma: Definition and laboratory investigation. ● Describe hemorrhagic disorders- classification, etiology, causes of gum bleeding, DIC, ITP, Investigation procedures (screening tests) ● Interpret, TC, DC, absolute values. PCV, Blood film, ESR, platelet counts, Bone marrow examination, BT, CT & PT. & determine clinical significance	B. Hematology (Disorders of blood) ● Introduction of haematology: Haemopoiesis, ● Anaemia, and classification & Aetiology of Anaemia, ● Common anaemias: Iron deficiency anaemia, megaloblastic, pancytopenia & aplastic anaemia, Haemolytic anaemia: Thalassaemia & Haemoglobinopathies. ● Laboratory investigations of anaemia including haemolytic anaemia ● White blood cell disorders- Leucocytosis, Leucopenia, lymphocytosis and leucomoid blood picture. ● Classification, aetiology, clinical features and laboratory diagnosis of leukaemia (Acute and Chronic). ● Multiple myeloma: Definition and laboratory investigation ● hemorrhagic disorders- classification, etiology, causes of gum bleeding, DIC, ITP, Investigation procedures (screening tests) ● Interpretation of TC, DC, absolute values. PCV, Blood film, ESR, platelet counts, Bone marrow examination, BT, CT & PT.	Lecture-16hrs Tutorial-20 hrs Practical-5 hrs

C-Clinical Pathology

Learning Objectives	Contents	Teaching Hours
Student will be able to : ● Define and classify of DM ● Describe hypoglycaemia and hyperglycaemia, Diagnose DM with OGTT, ● Describe complications of DM, Glycosuria and Benedict's test, importance of DM in dentistry ● Define jaundice & state classification of jaundice, ● Mention indication of LFT, and normal values of liver enzymes and bilirubin with their interpretation, ● Mention importance of jaundice in dentistry. ● Mention indications of RFT, and describe urine examination, proteinuria, haematuria and pyuria, heat coagulation test, Azotemia, uraemia, ● Mention biochemical values of urea and creatinine with their interpretation. ● Name electrolyte and lipid metabolic disorders and name major electrolytes with their normal values and units of measurements. ● State definition and components, indication and clinical importance of lipid profile.	C. Clinical Pathology ● Carbohydrate metabolic disorders: Define and classification of DM ● Hypoglycaemia and hyperglycaemia, Diagnosis of DM with OGTT ● Complications of DM, Glycosuria and Benedict's test, importance of DM in dentistry ● Liver functions test: jaundice, classification of jaundice, indication of LFT, normal values of liver enzymes and bilirubin with their interpretation, importance of jaundice in dentistry. ● Renal function test: Indication of RFT, Urine examination, proteinuria, haematuria and pyuria, Heat coagulation test, Azotemia, uraemia, Biochemical values of urea and creatinine with their interpretation. ● Electrolyte and lipid metabolic disorders: Major electrolytes with their normal values and units of measurements. ● Lipid profile: Definition and components, indication and clinical importance of lipid profile.	Lecture-10 hrs Tutorial-15hrs Practical-3 hrs

Microbiology

Learning Objectives	Contents	Teaching Hours
Student will be able to : ● Define & classify micro-organism-bacteria, virus & fungus. ● State bacterial, anatomy, physiology, growth & death. ● Define Sterilization, mention different methods & importance in dental practice. ● State normal flora – pathogenesis and host defense. Virulence factors of bacteria. ● State pathogenesis of dental caries. ● Classify Antimicrobial agents and state their mode of action and drug resistance. ● State culture media with sensitivity & its clinical importance. ● State (a) gram positive cocci - Staphylococcus, Streptococcus, Pnuemococci (b) gram positive bacilli- C.diphtheriae ,Clostradium species (c) gram negative cocci- Neisseria species (d) gram negative bacilli- Enterobacteriaceae .With common diseases they produce. ● State mycobacteria, actinomyces and their pathogenesis. ● Describe (a) Urine-R/E, aetiology of UTI and lab diagnosis (b) Stool R/E, aetiology of Diarrhoea and lab diagnosis (c) Lab tech of Gram staining and AFB staining (d) lab diagnosis of-. (1) Sore throat (2) STD (3) PUO.	1. Bacteriology ● Classification of micro-organism-bacteria, virus & fungus. ● Bacterial anatomy, physiology, growth & death. ● Sterilization, different methods of sterilization & importance in dental practice. ● Normal flora – pathogenesis and host defense. Virulence factors of bacteria. ● Pathogenesis of dental caries. ● Classification of antimicrobial agents and their mode of action and drug resistance. ● Culture media with sensitivity & its clinical importance. ● (a) Gram positive cocci - Staphylococcus, Streptococcus, Pnuemococci (b) gram positive bacilli- C. diphtheriae , Clostradium species (c) gram negative cocci- Neisseria species (d) gram negative bacilli- Enterobacteriaceae .With common diseases they produce. ● Mycobacteria, actinomyces and their pathogenesis. ● Clinical Microbiology – (a) Urine-R/E, aetiology of UTI and lab diagnosis (b) Stool R/E,aetiology of Diarrhoea and lab diagnosis (c) Lab tech of Gram staining and AFB staning (d) lab diagnosis of-. (1) Sore throat (2) STD (3) PUO (4) Different swab and pus	General bacteriology Lecture-10 hrs Systemic bacteriology lecture-20 hrs Tutorial-30 hrs Practical-8 hrs

Learning Objectives	Contents	Teaching Hours
Student will be able to : ● State the classification, structure, difference route of transmission of virus with examples. ● State hepatotropic viruses and describe their pathogenesis & lab. diagnosis ● Name important respiratory virus and state their pathogenesis, clinical importance & lab. Diagnosis ● Name enteric viruses and state Polio & Rota virus and also herpes family viruses. ● Demonstrate knowledge of new emerging virus – HIV, Birdflue, Nipa and Corona virus etc. Student will be able to : ● Demonstrate knowledge about general concept of fungus with clinical importance. State common fungal diseases of oral cavity (oral thrush) ● Classify according to site of infection and describe clinical presentation & lab diagnosis. ● State in short about Systemic & opportunistic funguses. Student will be able to : ● Define parasitology ● Classify Potozoa (broad classification),describe life cycle with lab diagnosis of–Ent-Histolytica, G-Intestanalis, Leish-donovani, malarial parasite ● Describe Life cycle & lab diagnosis of Nematodes –(a) Intestinal – Ascaris lumbricoides Ankylostoma doudenale, Trichuris trichiura,Enterobius vermicularis. (b) Tissue Nematodes – Wuchereria bancrofti. ● Mention short description of Cestodes and Trematode .	2. Viruses ● Introductory virology – General concept of virus, classification, structure, difference route of transmission with examples. ● Hepatotropic viruses their pathogenesis & lab. diagnosis ● Important respiratory virus with their pathogenesis, clinical importance & lab. Diagnosis ● Enteric viruses – Polio & Rota virus and also herpes family viruses. ● New emerging virus – HIV, Bird flue, Nipa and Corona virus etc. 3. Fungus ● Introduction and general concept of fungus with clinical importance. Common fungal diseases of oral cavity (oral thrush) ● Classification according to site of infection with clinical presentation & lab diagnosis. ● Systemic & opportunistic funguses 4. Parasitology ● Introduction of parasitology ● Potozoa (broad classification), life cycle with lab diagnosis of– Ent. Histolytica, G.Intestanalis, Leish-donovani, malarial parasite ● Life cycle & lab diagnosis of Nematodes –(a) Intestinal – Ascaris lumbricoides Ankylostoma doudenale,Trichuris trichiura, Enterobius vermicularis. (b) Tissue Nematodes: Wuchereria bancrofti. ● Cestodes and Trematode–short description	Lecture-8hrs Tutorial-6hrs Lecture-3hrs Tutorial-2hrs Lecture-10hrs Tutorial-5hrs Practical-3 hrs

Learning Objectives	Contents	Teaching Hours
Student will be able to : ● Demonstrate knowledge about general concepts of immunology and its applications, ● classify immunity ● Define – Antigen, immunogenic, hapten and antibody. ● Classification of antigen. ● State structure, types, functions of immunoglobulin & Ag-Ab reaction. ● Mention definition, sources & different pathways of complement. ● Mention definition, distribution & types with their functions of major histocompatibility complex (MHC)-. ● Describe formation & maturation of T & B Cell and their differences & functions ● Mention definition & classification of hypersensitivity and state mechanism & function. ● Demonstrate knowledge of introduction of Immunodeficiency, Immuno therapy, Autoimmunity & Tumour immunity.	Immunology ● General concepts of immunology and its applications, immunity – classification & difference. ● Antigen, immunogen, hapten and antibody. Classification of antigen. ● Immunoglobulin – structure, types, functions & Ag-Ab reaction. ● Complement – definition, sources, functions, pathways and their disorder. ● Major histocompatibility complex (MHC): Definition, distribution & types with their functions. ● Lymphatic traffic system – Formation & maturation of T & B Cell their differences & functions ● Hypersensitivity – Definition & classification with mechanism & function. ● Introduction of Immunodeficiency, Immuno therapy, Autoimmunity, Tumour immunity and transplant.	Lecture-8 hrs Tutorial-4 hrs